Copernicus Marine Service

QUALITY INFORMATION DOCUMENT

TAC SST Production Centre
SST_ATL_SST_L4_NRT_OBSERVATIONS_010_025

Issue: 3.0

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CHANGE RECORD

When the quality of the products changes, the QuID is updated and a row is added to this table. The third column specifies which sections or sub-sections have been updated. The fourth column should mention the version of the product to which the change applies.

Issue	Date	§	Description of Change	Author	Validated By
1.0	21/01/2019	All	Creation of the document First version for product 010-025 replacing product 010-003. Statistics for the product 010-025 over the European North West shelf/Iberia Biscay Irish Seas	Emmanuelle Autret	Mercator Ocean
1.1	26/03/2019	All	Review of the document	Emmanuelle Autret	Mercator Ocean
1.2	15/04/2019	All	Modified time period covered by the product (extended)	Emmanuelle Autret	Mercator Ocean
1.3	10/09/2019	All	EIS December 2019 Data from VIIRS instruments added to the input dataset	Emmanuelle Autret	Mercator Ocean
1.4	10/09/2020	All	EIS December 2020 Data from NPP, N20, Sentinel-3A & 3B added to the input datasets	Jean-François Piollé	Bruno Buongiorno Nardelli
2.0	16/06/2022	All	Adaptation to new template Update for EIS Nov 2022. Changed intercalibration methodology. Updated statistics	Jean-François Piollé Dominique Briand	A. Pisano
3.0	28/05/2025	All	Update for EIS Nov 2025 added NOAA-21 VIIRS as input sensor	Jean-François Piollé	A. Pisano

ACRONYM TABLE

Acronym	Description
Argo	International profiling float network (www.argo.net)
CMEMS	Copernicus Marine Service
EAN	Estimated Accuracy Number
GHRSST	Group for High Resolution Sea Surface Temperature
L2P	Level 2 Pre-processed
L3C	Level 3 Collated
L4	Level 4 (Analysed)
MY	Multi-year
NRT	Near Real Time
OSI SAF	Eumetsat Ocean and Sea Ice Satellite Application Facility
REP	Reprocessing
SST	Sea Surface Temperature
TAC	Copernicus Marine Service Thematic Assembly Centre

KEY DEFINITIONS

sea surface foundation temperature	**CF Definition**: The surface called "surface" means the lower boundary of the atmosphere. The sea surface foundation temperature is the water temperature that is not influenced by a thermally stratified layer of diurnal temperature variability (either by daytime warming or nocturnal cooling). The foundation temperature is named to indicate that it is the temperature from which the growth of the diurnal thermocline develops each day, noting that on some occasions with a deep mixed layer there is no clear foundation temperature in the surface layer. In general, sea surface foundation temperature will be similar to a night-time minimum or pre-dawn value at depths of between approximately 1 and 5 meters. In the absence of any diurnal signal, the foundation temperature is considered equivalent to the quantity with standard name sea_surface_subskin_temperature. The sea surface foundation temperature defines a level in the upper water column that varies in depth, space, and time depending on the local balance between thermal stratification and turbulent energy and is

	expected to change slowly over the course of a day. If possible, a data variable with the standard name sea_surface_foundation_temperature should be used with a scalar vertical coordinate variable to specify the depth of the foundation level. Sea surface foundation temperature is measured at the base of the diurnal thermocline or as close to the water surface as possible in the absence of thermal stratification. Only in situ contact thermometry is able to measure the sea surface foundation temperature. Analysis procedures must be used to estimate sea surface foundation temperature value from radiometric satellite measurements of the quantities with standard names sea_surface_skin_temperature and sea_surface_subskin_temperature. Sea surface foundation temperature provides a connection with the historical concept of a "bulk" sea surface temperature considered representative of the oceanic mixed layer temperature that is typically represented by any sea temperature measurement within the upper ocean over a depth range of 1 to approximately 20 meters. The general term, "bulk" sea surface temperature, has the standard name sea_surface_temperature with no associated vertical coordinate axis. Sea surface foundation temperature provides a more precise, well-defined quantity than "bulk" sea surface temperature and, consequently, is more representative of the mixed layer temperature. The temperature of sea water at a particular depth (other than the foundation level) should be reported using the standard name sea_water_temperature and, wherever possible, supplying a vertical coordinate axis or scalar coordinate variable.

RELEVANT DOCUMENT LIST

GHRSST GDS	The Recommended GHRSST Data Specification (GDS) Revision 2.1, June 2022, https://ghrsst.org
PUM	Related Product User Manual https://documentation.marine.copernicus.eu/PUM/CMEMS-SST-PUM-010-025.pdf

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I EXECUTIVE SUMMARY

I.1 Products covered by this document

This document describes the quality of the SST_ATL_SST_L4_NRT_OBSERVATIONS_010_025, a Copernicus Marine Service (CMEMS) SST TAC (Sea Surface Temperature - Thematic Assembly Centre) product. It is a multi-sensor level 4 (L4) analysis of Sea Surface Temperature, covering the European North West shelf/Iberia Biscay Irish Sea/Canary area (ATL) at 0.02° horizontal resolution, from 2021 to present. It is produced by Ifremer (Brest, France) production unit.

This dataset results from the merging of various satellite SST level 2 (L2) data, which have passed a significant number of quality controls and editing, and which have been inter-calibrated through an inter-sensor bias correction procedure, using a median field generated from a set of "best quality" sensors to provide an estimate of the night time SST, based on input SST observations and finally optimally interpolated.

I.2 Summary of the results

The baseline for the validation of SST TAC products is drifting buoy measurements, as currently recommended by the GHRSST[1] group on satellite SST validation (STVAL). The mean and standard deviation of the departures from drifting buoy SST measurements are computed over the ATL area (9°N - 62°N, 21°W - 13°E; see Table 2 below). Note that the validation results cover an 11-month period (01/02/2021 to 31/12/2021).

Integrated over the whole domain, the bias in surface temperature is -0.056°C with a standard deviation of 0.322°C.

I.3 Estimated Accuracy Numbers

This section includes the estimated accuracy numbers for the products covered in this document. The metrics applied are the following: mean (°C), standard deviation (°C), and number of match-ups for differences between High Resolution SST analysis over the ATL area (and sub areas: North-Western Shelves and Iberia Biscay Irish Sea) and in-situ SST measurements provided by the INS TAC. The statistics were computed for the dates between 2021-02-01 and 2021-12-31 (Table 1).

[1] http://www.ghrsst.org

Table 1: Estimated Accuracy Numbers. Metrics: mean, standard deviation and number of match-ups for differences between High Resolution SST analysis over the whole ATL area and subareas European North West Shelf and Iberia Biscay Irish Sea (IBI ROOS) area and in-situ SST measurements provided by the INS TAC. Period: 01-02-2021 and 31-12-2021.

Area	**Mean (°C)**	**Standard deviation (°C)**	**Number of matchups**
ATL (8°N to 62°N, 21°W to 13°E)	-0.056 ± 0.001	0.322 ± 0.001	74273
North Western Shelves (48°N to 62°N, 18°W to 13°E)	-0.081 ± 0.001	0.299 ± 0.001	11735
IBI ROOS (28°N to 56°N, 19°W to 5°E)	-0.058 ± 0.001	0.309 ± 0.001	47729

II PRODUCTION SYSTEM DESCRIPTION

II.1 Production centre description

The main properties of this dataset are provided in Table2 below.

Table 2: Properties of the product.

Production centre name	**IFREMER, FR**
Product name	SST_ATL_PHY_L3S_NRT_OBSERVATIONS_010_025
Short description	L4 daily maps of SST over North East Atlantic
Variables/datasets	sea_surface_foundation_temperature
Reprocessing	No
Geographical coverage	North East Atlantic (ATL): North-West Shelf, Iberia, Biscay Irish Sea, Canary Islands: 9°N - 62°N, 21°W - 13°E
Spatial resolution	0.02 x 0.02° (equirectangular grid)
Vertical coverage	Surface
Temporal coverage	2018-01-01 – present
Temporal resolution	Daily
Target delivery time	Daily at 12pm (noon) UTC for day-1 00:00 UTC Analysis
Update Frequency	Daily

II.2 Description

The European North West shelf/Iberia Biscay Irish Sea/Canary area (ATL) (shown in Figure 1) – High Resolution L4 Sea Surface Temperature Analysis was produced with the Ifremer ODYSSEA Sea Surface Temperature L4 processing chain, currently using (as of June 2025) version 3.0.

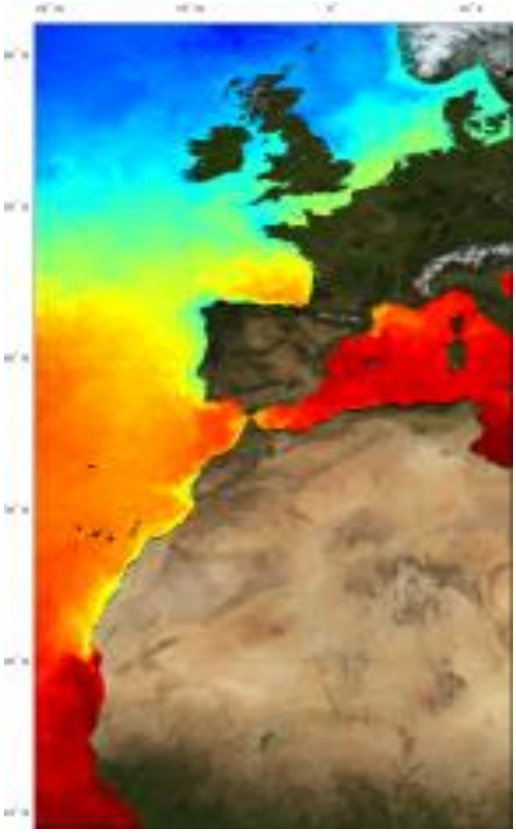

Figure 1: ATL area covered by this product.

Figure 2 shows the different steps followed for processing ODYSSEA SST products at Ifremer. The different steps are detailed hereafter.

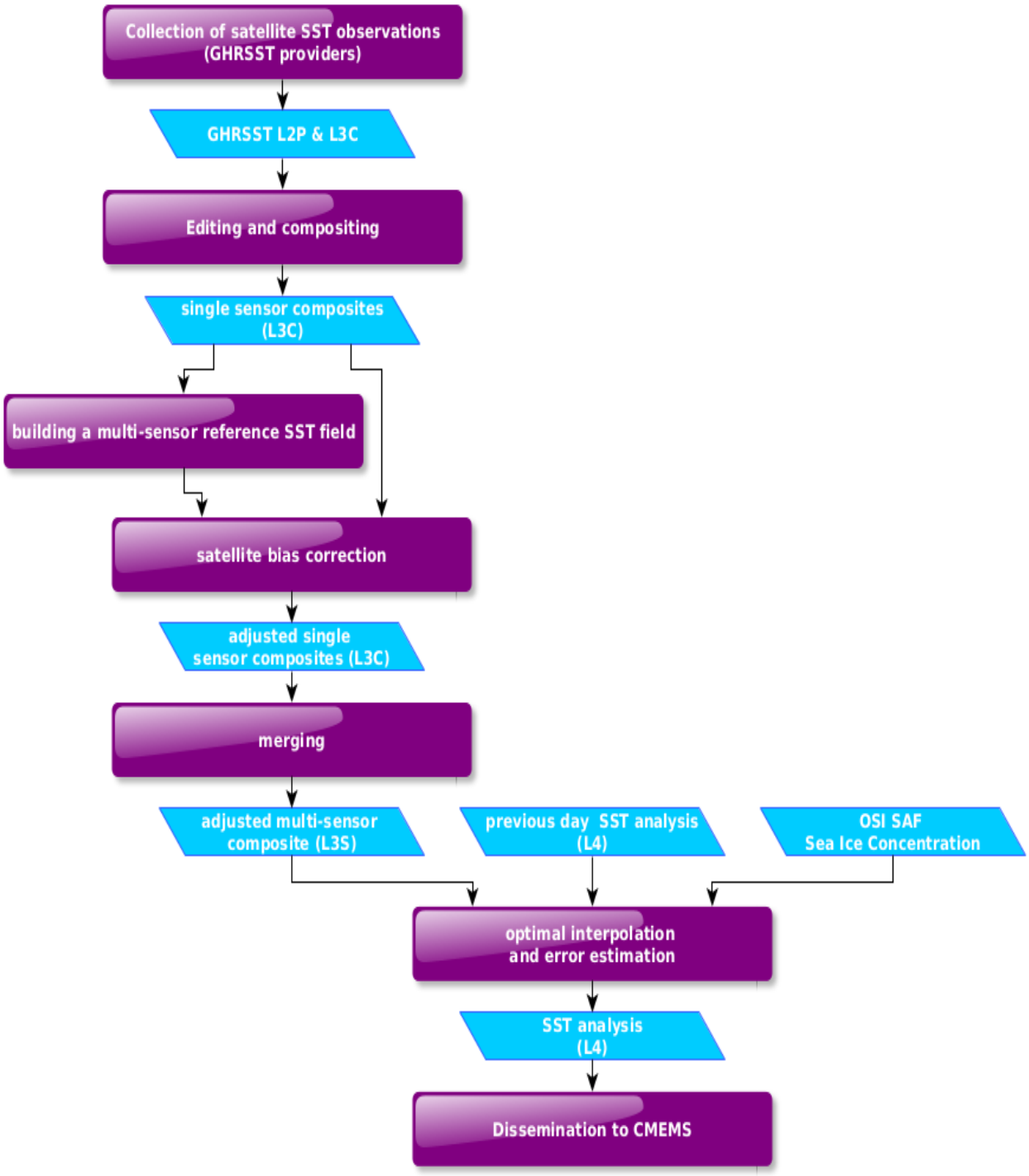

Figure 2: Schematic diagram of the ODYSSEA operational processing chain at IFREMER.

II.3 Collection of inputs

In the current configuration of the processing chain, a selection of GHRSST L2P (Level 2 pre-processed, for polar orbiting satellites) and L3C (Level 3 Collated, for geostationary satellites) products from different producers (NASA, NOAA, EUMETSAT OSI-SAF, ESA) - in GHRSST compliant format [R-1] - are collected, that are merged into the final product.

All L2P/L3C data are provided with Sensor Specific Error Statistics (SSES_bias and SSES_std variables) that give an estimate of the systematic and random errors at pixel level. In addition, quality level flags are provided.

The L2P data collected currently or for older dates back to 2018 for ODYSSEA are listed in the table below (table 1).

Table 3: Satellite L2P data collected for ODYSSEA. In yellow the L2P currently ingested as of June 2025, others sources (in white) were used in previous years only, depending on what satellite missions were available at that time.

Satellite input data	Provider	Resolution	Used in intercalibration reference	Status
AVHRR GAC NOAA17, 18,19	NAVO	9 km	Yes	Missions terminated
AVHRR LAC NOAA17,18,19	NAVO	1 km	Yes	Missions terminated
AATSR ENVISAT	ESA	1 km	Yes	Mission terminated
AVHRR METOP-B	OSI SAF	1 km	Yes	
SEVIRI MSG	OSI SAF	5 km / 3hr	Yes	
AVHRR NAR NOAA17/18/19	OSI SAF	2 km	Yes	Missions terminated
AVHRR NAR / METOP-A	OSI SAF	1 km	Yes	Mission terminated
MODIS / AQUA	PODAAC	1 km	No	Replaced with other sources
NPP / VIIRS	OSPO	1 km	Yes	
NOAA-20 / VIIRS	OSPO	1 km	Yes	
NOAA-21 / VIIRS	OSPO	1km	Yes	
AMSR-E	REMSS	30 km	Yes	Mission terminated
AMSR-2	REMSS	30 km	No	
TMI	REMSS	30 km	Yes	Mission terminated
Sentinel-3 A / SLSTR	EUMETSAT	1 km	Yes	
Sentinel-3 B / SLSTR	EUMETSAT	1 km	Yes	

In addition, the following complementary datasets are used:

- **Sea-ice concentration** data from the EUMETSAT OSI-SAF[2] are used in ODYSSEA to mask out sea ice covered areas.
- **Climatology**: the SST climatology used has been derived from the PATHFINDER SST 5-daily climatology produced by Casey and Cornillon (1999) [R.2]. The fields have been spatially smoothed to reduce possible artefacts due to small scales and interpolated in time (daily).

II.3 Editing and compositing

All satellite sea surface temperature (SST) data valid for a particular day (from previous day 12:00 to current day 12:00) are extracted from the downloaded L2P/L3C files and regridded into a single sensor composite file at the final resolution grid, for each region to process (i.e., ODYSSEA products datasets at global scale and on specific regions including this one). It means that a high-resolution file (< 10 km) will be processed at least two times, once for the global high-resolution grid (10 km) and once for the ultra-high resolution regional grids (2 km). The input SST data undergo various editing and processing steps:

- only satellite data which have a quality level (provided in each GHRSST compliant dataset [R.1]) above a certain value (which may be different for each satellite input and linked to GHRSST GDS) are accepted. This minimum quality level can be different depending on the data source and is adjusted per source after expert evaluation.
- day-time data (determined using a solar zenith angle calculation) are rejected. Only nighttime data are kept to provide an SST free of diurnal cycle close to the foundation value (see Key definitions table above).
- the SSES biases supplied with the GHRSST data are removed from each pixel and the SSES standard deviation values are passed on to the next steps in the analysis chain.
- Some SST provider calibrate their SST to skin layer (first micrometers), other to subskin (first millimeter): in order to harmonize all input SST, for skin temperature datasets (e.g., SLSTR on Sentinel-3A and Sentinel-3B), a skin-to-subskin correction is applied [R.3]. Measurements where wind speed is lower than 2 m/s are ignored to avoid night time cooling at skin/subskin layer.
- incidence angle for geostationary sensors is also taken into account to remove pixels with too high incidence (greater than 55°).
- ice concentration (only at high latitudes when ice mask from OSI SAF is applied) in the L2P/L3C is used to remove pixels with ice contamination greater than 10%
- aerosols, when provided in the L2P /L3C is used to remove pixels with aerosols content greater than 0.3
- various filters may be applied too, depending on the SST source (valid range for SST, type of SST retrieval algorithm, etc.)
- pixels having the highest quality and falling into the same target grid cells are averaged together provided that a minimum number of clear-sky pixels is available in order to avoid unrepresentative grid cells. The number of pixels varies with respect to the sensor native resolution and final grid resolution, as part of the processing chain internal configuration and therefore customized based on internal input assessment for each source.

[2] https://osi-saf.eumetsat.int/

II.4 Bias correction

Satellite data can be biased for several reasons, including:

- atmospheric water vapour;
- atmospheric aerosol (dust);
- surface changes (e.g., extreme roughness);
- instrument calibration problems.

These effects can lead to biases with regards to in situ measurements and spatio-temporal discrepancies in the analysis if they are not addressed. In order to prepare a consistent daily dataset, an intercalibration method has been developed. This processing step consists in estimating a daily correction to be applied on SST values for each single satellite source. The main issue is to define a daily reference with which to estimate this correction. Then for each single source, a large-scale bias field (with synoptic weather scales) is estimated from the differences between the observations and the reference of the day, and the estimated correction is applied on the SST values of each satellite source.

The first version of ODYSSEA only used the Envisat/AATSR observations as a reference and daily regional biases were estimated from ten days of match-ups. This method gives good results on average but presents the disadvantages of depending on one instrument and on the sampling of the match-ups. The method adopted in the second version of ODYSSEA for Medspiration and Copernicus consists in constructing a daily reference field (on a 0.25° grid) by optimal interpolation (Eq. 1) from a dedicated dataset including all the sources. It was however noted that this method introduced a seasonal bias. The current methodology, implemented as of November 2022, involves building a reference SST field (on a 0.10° grid) combining the measurements of the “best source” (currently VIIRS onboard NPP as processed by NOAA/OSPO) with the median value of a selection of trusted sources (refer to Table 3 for the list of products currently taken into account in this median calculation). The choice of the best source and the trusted sources for the median calculation was evaluated from comparisons with in situ measurements over a year of data.

The difference between each source to be merged into the final product and the computed reference is then smoothed with a large scale gaussian filter and removed from the original SST to provide an **adjusted sea surface temperature**.

The outcome of this process is a new version of the single composite satellite data (L3C), which has been bias-corrected (or adjusted).

II.5 Merging

The daily single sensor composite files are then merged together into a combined multi-sensor composite file (L3S) at the final product resolution. The single sensor composites are first regridded, if necessary, on the final grid resolution (using bilinear interpolation or averaging depending on whether the single sensor is coarser or has higher resolution than the final grid). The merging process then takes into account a hierarchy of sensors (from best to worst, based on an internal assessment again from comparisons of each source with in situ measurements, performance of cloud masking, coverage. A given cell of the final grid will contain a value from one source only (the best selected). The used source is reported in the *sources_of_sst* variable of the created L3S.

II.6 Optimal interpolation and error estimation

We use the state space model representation

$$x^t(t_i) = x(t_{i-1}) + B^{1/2}\eta(t_i \qquad (1)$$

$$y^o(t_i) = Hx^b(t_{i-1}) + R^{1/2}\varepsilon^o(t_i) \qquad (2)$$

With $y^o(t_i)$ observations at time t_i , $x^t(t_i)$ hidden state vector (true SST measured by $y^o(t_{i)}$ at time t_i), *H* the observation operator corresponding to a possible grid transformation (model grid to observation grid), *R* the covariance error matrix between state and observations (the diagonal elements are the measurement errors, e.g. SSES standard deviation errors in L2P files), B the covariance error matrix empirically computed (with one year of METOP AVHRR data)

Variograms are fitted by an exponential isotropic model (Eq. 3).

$$C(d) = \sigma^2 \exp(-\frac{d^2}{2L^2}) \qquad (3)$$

with d the distance between two points and L the regional length scale. Note that the correlation scales are of the order of Rossby Radius (bounded by 10 km and 100 km). The optimal solution is given by optimal interpolation.

The previous day global analysis is used as background field. For spin-up or if the analysis is not available for more than 7 days, the analysis uses a reference daily climatology. The 4 km Pathfinder Version 5 (5-day) climatology was used to construct this reference, smoothed and interpolated and remapped to daily and 10 km time and space resolution.

II.7 Ice and land mask

A land mask has been defined on the ODYSSEA v1.0 analysis grid. The land mask was extracted from GMT (Wessel et al. 2006).

A sea-ice mask is defined for each analysis in high latitudes. In the configuration of the daily system ODYSSEA, the ice mask is defined daily by using the ice product provided by OSI SAF. The variable *sea_ice_fraction* provided in this product is remapped on the analysis grid and copied in our L4 file. However, for each analysis the ice data is extrapolated (with the nearest valid value) to fill the icy regions (with sea pixels) close to land.

The ice mask and the land mask used to mask out the SST in the L4 product are stored in the variable *mask*. The variable *sea_ice_fraction* resampling is based on on L4 grid of the sea ice fraction provided by the OSI SAF product.

III VALIDATION FRAMEWORK

The baseline for the validation of SST TAC products is only based on drifting buoy measurements, as currently recommended by the GHRSST group on satellite SST validation (STVAL). The mean and standard deviation of the departures from drifting buoy SST measurements are computed over the whole ATL area and North West shelf/Iberia Biscay Irish Sea/Canary area regions. Various quality check procedures are applied on drifting buoy measurements to discard suspect data, using gross error checks and buoy blacklists (identifiers of bad buoys) based on intercomparison with satellite made available by UK MetOffice and Meteo-France/CMS. Only night-time measurements are kept, to be consistent with the data screening of the input satellite data.

Overall, the L4 SST product over ATL region is purely based on satellite observations only, and hence drifting buoy measurements can be considered to be a fully independent validation data. However, in some regional seas, the number of drifting buoy measurements is very low (e.g., Baltic Sea), and even equal to zero. In the future, other data sources (moored buoys, ARGO floats, etc.) could be considered for validation statistics if available.

The present validation study specifically uses drifting buoy measurements provided by the CMEMS IN-SITU TAC (INSITU_GLO_PHYBGCWAV_DISCRETE_MYNRT_013_030[3]). The analysis minus observation differences is calculated by averaging first the in situ data over the same grid cell as the satellite dataset. The mean and standard deviation of these differences are then calculated over the whole time series for each grid cell.

These diagnostics are processed for every day of the time series and over the validation year, for the full ATL domain and for the different sub-regions.

[3] https://data.marine.copernicus.eu/viewer/expert?view=layers&dataset=INSITU_GLO_PHYBGCWAV_DISCRETE_MYNRT_013_030

IV VALIDATION RESULTS

Table 4 shows the bias, standard deviation and number of observations of the ATL L4 observations minus drifting buoy differences for the whole ATL domain and two validation sub-regions (North Western Shelves and IBI ROOS) for the period 01/02/2021 to 31/12/2021.

Table 4: Mean, standard deviation and number of match-ups for differences between High Resolution SST analysis over the European North West Shelf/Iberia Biscay Irish Seas (IBI ROOS) and in-situ SST measurements provided by the INS TAC. The statistics were computed for the dates between 2021-02-01 and 2021-12-31.

Area	Mean (°C)	Standard deviation (°C)	Number of samples
ATL (8°N to 62°N, 21°W to 13°E)	-0.056 ± 0.001	0.322 ± 0.001	99916
North Western Shelves (48°N to 62°N, 18°W to 13°E)	-0.081 ± 0.001	0.299 ± 0.001	40708
IBI ROOS (28°N to 56°N, 19°W to 5°E)	-0.058 ± 0.001	0.309 ± 0.001	53602

A visual breakdown of some of the statistical metrics, mean bias and number of matchups is shown in Figure 2 for the same time period (2021-02-01 to 2021-12-31). This shows the variation of the mean bias and number of matchups over the area in the validation period.

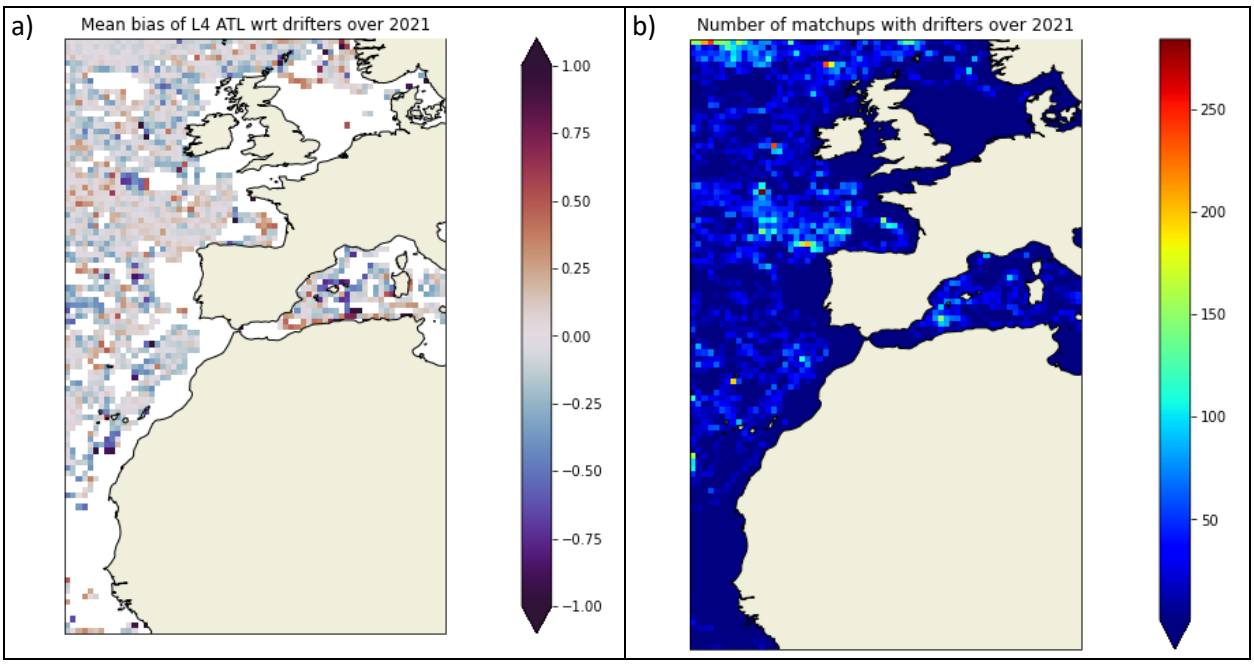

Figure 2: Spatial distribution of (a) mean bias (C°) and (b) number of available match-ups for differences computed over the period from 2020-02-01 to 2020-01-31 (white areas correspond to no data)

V SYSTEM'S NOTICEABLE EVENTS, OUTAGES OR CHANGES

This section is dedicated to describe and track changes that may occur in the current operational system. It includes changes due to outages (data gaps), upstream data (e.g., addition or loss of a sensor, inclusion/removal of in-situ platforms), reprocessing, corrections, assimilated data, boundary conditions, among other changes that may impact product quality.

Date	Change/Event description	System version	Other
21/03/2019	SST_ATL_SST_L4_NRT_OBSERVATIONS_010_025 is a new product. Compared to the SST_NWS_SST_L4_NRT_OBSERVATIONS_010_003 product, the change corresponds to an increase in the extent of coverage area to the west of the Azores and the Canary Islands to provide a more complete product.	2.0	
10/09/2019	SST data from VIIRS instruments are included in the input dataset.	2.0	
December 2020	Data from NPP, N20, Sentinel-3A and S3B added to the input datasets MODIS, NOA19 removed from input datasets.	2.0	
November 2022	Revision of the intercalibration across the different input missions to reduce seasonal bias	3.0	
November 2025	Data from NOAA-21/VIIRS added to the input datasets	3.0	

VI QUALITY CHANGES SINCE PREVIOUS VERSION

This section aims to provide a short summary of the main changes in Product Quality between the previous and current version.

The previous version was released in November 2022. This update covers the addition of NOAA-21/VIIRS in the list of input streams to this multi mission dataset, to anticipate the decommissioning of NOAA-NPP/VIIRS. The sensor and processing for these two inputs are identical (and provided by NOAA).

The quality of the dataset remains within the same range as the previous release (absolute bias lower than 0.1° and standard deviation about 0.3 C°). For assessment, the product was generated over a period of three months (Oct to Dec 2024) with and without NOAA-21 VIIRS. With a mean difference of -0.02 C° and RMSD of 0.087 C°. It was concluded that the addition of NOAA-21 does not impact the product quality and the validation results remain consistent (i.e., comparisons with in situ data showed similar results, with similar bias and RMSD over the tested period with and without NOAA-21). The addition of this new data source increases on average by about 8% the number of observations in the L3S dataset which is significant in terms of additional coverage. However, this additional coverage will be only temporary, due to the upcoming decommissioning of NOAA-NPP (expected 2028). The use of NOAA-21 also improves the robustness of the L4 interpolation and dataset production, as NOAA-21 provides backup in case one or more of the other mission input datasets become unavailable (e.g., satellite man oeuvre, sensor malfunction, data provision issue, etc.).

VII REFERENCES

[R.1] The GHRSST Data Processing Specification (GDS) provides a detailed technical specification for producing GHRSST L2, L3 and L4 products. The latest version of this document is the GDS 2.1 version 0, published in summer 2022. The Group for High Resolution Sea Surface Temperature Science Team, Piollé, Jean-François, Armstrong, Ed, Casey, Kenneth, & Donlon, Craig. (2022). The Recommended GHRSST Data Specification (GDS) (GDS 2.1 revision 0). Zenodo. https://doi.org/10.5281/zenodo.6984989

[R.2] Wessel, P., and W. H. F. Smith, New, improved version of Generic Mapping Tools released, EOS Trans.Amer. Geophys. U., vol. 79 (47), pp. 579, 1998

[R.3] Donlon, C. J., P. J. Minnett, C. Gentemann, T. J. Nightingale, I. J. Barton, B. Ward, and M. J. Murray, 2002: Toward Improved Validation of Satellite Sea Surface Skin Temperature Measurements for Climate Research. J. Climate, 15, 353–369, https://doi.org/10.1175/1520-0442(2002)015<0353:TIVOSS>2.0.CO;2